

Why Important Questions Are Often Difficult

Significance, Persistence, and the Composition of Unresolved Inquiry

Honggao Cao

September 2026

Abstract

Important questions often appear unusually difficult, but significance, difficulty, and non-resolution are distinct properties. This paper asks how significant and difficult questions become prominent among inquiries that remain active and unresolved. The analysis separates two exits from active inquiry: resolution and disengagement. Greater question difficulty can lower the rate of resolution, while greater question significance can lower the rate of disengagement. These processes can enrich the surviving population in both difficult and significant questions. A competing-risks representation shows that such enrichment does not by itself create statistical dependence when significance and difficulty are independent at entry and the exit processes are separable. Statistical association requires prior dependence, non-separable selection, or cross-effects between the two exit processes. The framework provides a diagnostic account of how populations of unresolved questions are formed and why the questions that remain visible can be both consequential and demanding.

Keywords: research questions; significance; objective difficulty; competing risks; research persistence; disengagement; scientific problem choice

1. Introduction

Some of the questions that matter most to us are also among the questions that remain hardest to settle. Questions about serious diseases can remain open after decades of research. Longstanding mathematical questions can survive generations of expert effort. Questions about valuable opportunities can change as others pursue them. Questions about fast-moving technologies can become easier in one respect while new questions emerge in another.

This paper explains how differential resolution and disengagement can enrich the population of active unresolved questions in both significance and difficulty, and identifies the additional conditions required for those dimensions to become statistically associated.

An active unresolved question can leave because it is resolved, or because meaningful pursuit stops. Difficulty bears most directly on the first exit. Significance bears most directly on the second.

Difficulty can keep a question from being resolved; significance can keep it from being dropped.

Consider two equally difficult questions, one highly consequential and one trivial. Neither is readily resolved, but sustained effort is more likely on the consequential question. Now consider two equally consequential questions, one relatively tractable and one difficult. The tractable question is more likely to leave through resolution. Repeated across many questions, these two filters change the composition of what remains visible.

The paper first defines the question-level unit and the two exit processes, then uses selected dynamics and related literature to locate the argument. A formal competing-risks section shows how survival reweights the unresolved population and distinguishes enrichment in significance and difficulty from statistical dependence between them.

2. Problems, Questions, and the Unit of Inquiry

Problems and questions are related but not identical. A problem is a broader condition, challenge, opportunity, or domain of concern. It can generate many questions, some already settled and others still open. The questions are the units of inquiry followed in this paper.

This distinction matters because a problem can remain important after particular questions about it have been answered. Chronic kidney disease remains consequential even though many questions about its causes, diagnosis, and management have established answers. Conversely, a specific question can have high scientific significance even when the broader practical problem is limited.

Problems provide contexts of concern; questions are what inquiry attempts to resolve.

The model operates at the level of questions. Problems or domains generate question populations but are not themselves the units followed through resolution and disengagement.

2.1 Question significance

For question i , significance refers to the consequence or value of resolving that question. It may reflect decision value, scientific value, practical consequence, systemic consequence, or value for managing the broader problem. Question significance may depend partly on the significance of the underlying problem, but the two are not identical.

2.2 Objective question difficulty

Objective difficulty refers to resistance to adequate resolution under the inquiry environment at time t . Relevant features may include available evidence, methods, tools, expertise, coordination, and target stability. Difficulty is therefore relative to the current inquiry environment rather than an intrinsic, time-invariant property of the question.

2.3 Unresolved status

Non-resolution is different again. A question may remain open because it resists sustained competent inquiry, or because it receives little serious attention. Work on undone science documents socially valuable research questions left incomplete or unstudied because research agendas are shaped by funding, institutions, and power [2].

Unresolved does not imply difficult.

3. The Population of Active Unresolved Questions

Consider a broad population of possible questions. Some never receive sustained attention. Some become active objects of inquiry. Once active, a question may be resolved, abandoned or deprioritized, transformed

into a different question, or revived after a period of dormancy. Resolution can also expose successor questions. Dynamic scientific problem choice and successor generation already have close precedent, including Carayol and Dalle's model of sequential problem choice in open science [1].

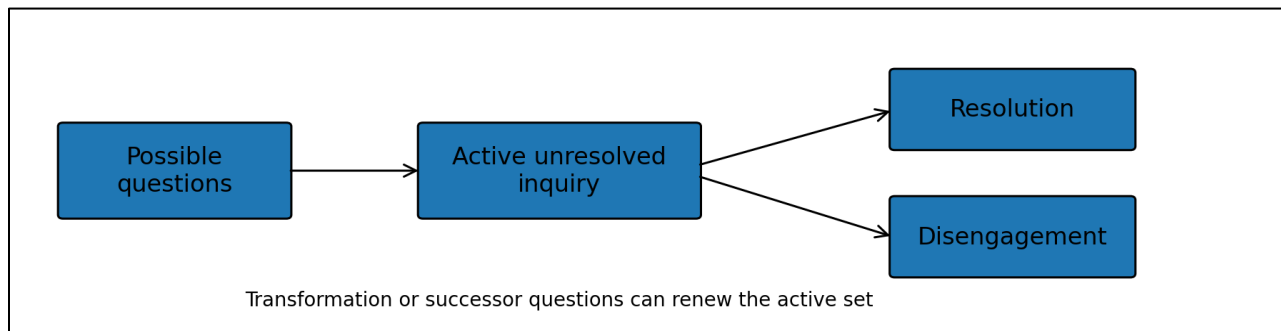
At the simplest level, the flow is: possible question -> active inquiry -> resolution or disengagement -> surviving unresolved population.

The questions still being actively pursued therefore form a survivor population.

The selection is intuitive. Easier questions tend to leave when answered; questions judged not worth continued effort can leave when pursuit stops. The remaining population reflects both filters.

Resolution and disengagement convey different information. Resolution indicates that inquiry reached an adequate answer under the relevant standard. Disengagement can reflect value, cost, institutional conditions, or changing priorities.

Figure 1. Entry, Active Inquiry, and Two Principal Exits



A question can disappear because it was answered or because people stopped trying to answer it.

3.1 Question difficulty and population difficulty

The difficulty of a particular question is not the same as the difficulty of the remaining population. A field can resolve many questions while the questions still under active inquiry remain demanding. Tractable questions leave through resolution; progress can reveal deeper questions; adaptive targets can change while inquiry proceeds. Research effort can also rise while research productivity falls in some domains, providing one empirical precedent for this distinction [5].

4. Two Exit Processes

Two exit processes jointly shape the composition of unresolved questions. Higher objective difficulty can reduce the rate of resolution. Greater significance can reduce the rate of disengagement. If both tendencies operate, difficult questions are retained because they are slow to resolve, while significant questions are retained because they continue to warrant effort.

Higher objective difficulty can reduce the rate of resolution. Greater significance can reduce the rate of disengagement. If both tendencies operate, difficult questions are retained because they are slow to resolve, while significant questions are retained because they continue to warrant effort.

Table 1 summarizes the directional pressures.

Table 1. Significance, Difficulty, and Typical Exit Pressure

	More tractable	More difficult
Lower significance	May be resolved cheaply or receive little attention	May remain technically open but lose active pursuit
Higher significance	Likely to attract effort and leave through resolution	Likely to remain visible while resolution is slow

These processes can make both high significance and high difficulty more prevalent among surviving questions.

Significance and difficulty can also affect both exits through additional channels. Significance may mobilize resources that raise the resolution rate; difficulty may influence persistence when challenging questions carry prestige or when expected attainability falls. Those cross-effects matter for the statistical relationship between the two dimensions.

5. Resolution Constraints

The exit structure explains how the population changes, but not why a particular question has a low resolution rate. In some cases that reason can be independently identified.

A causal question may lack a necessary observational contrast. A proof may depend on a method or intermediate result that can be specified independently of the fact that the question remains unresolved.

A resolution constraint counts as explanatory only when it can be specified independently of the observed non-resolution.

Information is one possible constraint. Methods, computation, instrumentation, coordination, and target stability can also limit resolution.

6. Selected Dynamics

6.1 Disease

Chronic kidney disease illustrates why an important problem should not be treated as one unresolved object. The disease remains a major health concern, while many questions about its causes, diagnosis, and management have established answers. The National Institute of Diabetes and Digestive and Kidney Diseases identifies diabetes and high blood pressure as the most common causes in adults and describes established approaches to diagnosis and management [7].

Other questions remain open. Their persistence can reflect technical difficulty, incomplete evidence, or research allocation. The relevant population is therefore a set of questions arising within a disease domain. Questions that become tractable can leave through resolution; consequential questions that remain difficult can continue to attract inquiry.

6.2 Longstanding mathematics

The Riemann hypothesis provides a stable-target question. Formulated in 1859, it remains unresolved despite sustained mathematical attention [8]. Its recognized scientific significance helps sustain attention, while its long history of unsuccessful resolution provides evidence of substantial technical difficulty.

Persistence is especially informative in this setting because the target is stable and the history of serious inquiry is substantial. The case illustrates how a highly significant question can remain active for a long period when disengagement is unattractive and resolution remains difficult.

6.3 Markets

Markets illustrate a setting in which the object of inquiry can change while it is being investigated or acted upon. Consider the question of whether a profitable strategy or opportunity can continue to generate excess returns. Rosenbaum's industry-level evidence finds that net entry responds positively to initial profits and negatively to entry barriers [9]. This supports the narrower point needed here: profitable conditions can attract entry, while barriers shape how strongly that response occurs.

The active question is whether an opportunity remains exploitable under changing competitive conditions. Its difficulty can rise as competitors enter, imitate, or adapt. Where entry is constrained by scarce expertise, regulation, network effects, capital requirements, or information advantages, the same question can have a different trajectory.

6.4 Artificial intelligence

Artificial intelligence illustrates a domain in which both inquiry capacity and the questions under inquiry can change quickly. A benchmark may initially provide a useful question about comparative capability, then lose discriminating power as systems improve. A governance or safety question can also change as new capabilities enable new uses [10].

Better tools and methods can make a fixed question easier to answer. At the same time, capability growth can generate successor questions or alter the conditions under which an existing question must be answered. The unresolved population can therefore renew itself even as individual questions are resolved.

6.5 What the selected dynamics show

Together, the cases show distinct ways in which the composition of unresolved questions can change: prior resolution, sustained persistence, adaptive response, capability growth, and successor generation.

7. Related Literature

7.1 Problem choice and search

Research on problem choice explains how questions become objects of inquiry. Carayol and Dalle [1] model sequential scientific problem choice and successor generation within an evolving knowledge structure. Problemistic-search research treats problems as triggers for search and examines how organizations respond when performance falls below aspirations [3]. Rzhetsky et al. [11] explicitly represent scientific opportunities in terms of importance and difficulty when studying experiment choice, and Alon [12] similarly frames

problem selection around interest and feasibility. These literatures establish that inquiry populations are shaped before any exit occurs.

7.2 Persistence, disengagement, and abandonment

A second literature concerns whether pursuit continues. Psychology distinguishes persistence from disengagement in goal pursuit [4]. Undone-science research shows that socially important questions can remain incomplete because research agendas are institutionally structured [2]. Operations research has modeled when an uncertain research project should be abandoned and replaced by another opportunity [13]. Innovation research likewise studies continuation, delay, abandonment, and completion within project portfolios [14], [15].

These strands make disengagement a substantive research object rather than a residual category. They also show that leaving an unresolved question can be rational, institutional, or resource-driven.

7.3 Selection and population composition

Selection changes what is observed after entry and exit. Endogenous-selection research formalizes how conditioning on a selected population can alter apparent relationships among variables [6]. R&D portfolio research studies ranking, continuation, cancellation, and selection across projects [16]. The present framework applies this population logic to unresolved questions by distinguishing two exits that can be influenced by different attributes.

7.4 Changing populations of unresolved questions

A further literature explains why progress need not produce an easier remaining set. Bloom et al. [5] document settings in which research effort rises while research productivity falls. Sequential problem-choice models allow resolved work to generate successor questions [1]. Adaptive domains add another channel: inquiry or exploitation can change the target itself.

Taken together, these literatures explain entry, continued pursuit, exit, selection, and renewal. The question developed here is what those processes imply jointly for the significance and difficulty composition of the questions that remain active and unresolved.

8. Formal Representation

8.1 Two ways a question can leave active inquiry

An unresolved question can leave active inquiry by being resolved or by being dropped. The formal model treats these as two competing exit processes. The purpose of the mathematics is to track how different exit rates reweight the mix of questions that remains.

Let $i=1,\dots,n$ index questions that enter active inquiry. Let T_i be the time from entry to the first exit from the active unresolved state, and let J_i in $\{R,A\}$ denote the exit type: resolution R or disengagement A . Let S_i denote the significance of resolving question i and $D_i(t)$ its objective difficulty at time t .

8.2 Representing the rates of exit

Define the cause-specific hazards

$$\lambda_R(t | S_i, D_i, X_i) = \lim_{h \rightarrow 0} \frac{\Pr(t \leq T_i < t + h, J_i = R | T_i \geq t, S_i, D_i(t), X_i(t))}{h}$$

$$\lambda_A(t | S_i, D_i, Z_i) = \lim_{h \rightarrow 0} \frac{\Pr(t \leq T_i < t + h, J_i = A | T_i \geq t, S_i, D_i(t), Z_i(t))}{h}$$

These equations separate the instantaneous rate of being resolved from the instantaneous rate of being dropped.

A proportional specification can be written as

$$\lambda_R(t | \cdot) = \lambda_{R0}(t) \exp\{-\beta_D D_i(t) + \theta_S S_i + \beta^\top X_i(t)\}$$

$$\lambda_A(t | \cdot) = \lambda_{A0}(t) \exp\{-\gamma_S S_i + \kappa_D D_i(t) + \gamma^\top Z_i(t)\}$$

Here $\beta_D > 0$ corresponds to lower resolution rates for harder questions, while $\gamma_S > 0$ corresponds to lower disengagement rates for more significant questions. The cross-terms θ_S and κ_D allow significance to affect resolution and difficulty to affect persistence.

8.3 How survival changes what remains

The total exit hazard is

$$\lambda(t | s, d) = \lambda_R(t | s, d) + \lambda_A(t | s, d)$$

and the survival probability in the active unresolved state is

$$G(t | s, d) = \exp\left\{-\int_0^t [\lambda_R(u | s, d) + \lambda_A(u | s, d)] du\right\}$$

$G(t|s,d)$ is the probability that a question of type (s,d) is still active and unresolved at time t .

Let $f_0(s,d)$ be the joint density of significance and difficulty at entry. Among surviving questions at time t ,

$$f_t(s, d) = \frac{f_0(s, d)G(t | s, d)}{\int \int f_0(a, b)G(t | a, b) da db}$$

The unresolved population is therefore the entry population reweighted by how long each type of question tends to remain active.

8.4 Enrichment is not the same as correlation

The distinction is easiest to see with a small example. Suppose equal numbers of questions enter in four groups, as shown in Table 2.

Table 2. Entry Counts and Separable Survivor Weights

	Entry: easier	Entry: harder	Survivor: easier	Survivor: harder
Lower significance	25	25	4	8
Higher significance	25	25	8	16

Suppose harder questions are twice as likely to remain because they resolve more slowly, while more significant questions are twice as likely to remain because they are less often abandoned. The survivor weights in Table 2 apply these two filters independently.

The remaining population is richer in hard questions, richer in significant questions, and contains many more high-significance/high-difficulty questions than before. Yet the 4:8:8:16 survivor pattern still factorizes exactly, so significance and difficulty remain statistically independent.

The same result appears algebraically. Consider the special case $\theta_S = \kappa_D = 0$, so significance affects disengagement and difficulty affects resolution through separate channels. If S and D are also independent at entry, then

$$\lambda_R(t | s, d) = r_t(d), \quad \lambda_A(t | s, d) = a_t(s)$$

then

$$G(t | s, d) = G_D(t | d) G_S(t | s)$$

and the reweighted density factorizes. Factorization means that the joint survivor distribution can be written as the product of its significance and difficulty components; the two dimensions therefore remain statistically independent.

Separate filters can enrich both dimensions without creating dependence between them.

A statistical relationship between significance and difficulty requires additional coupling: dependence already present at entry, significance changing the resolution rate, difficulty changing disengagement, explicit S-D interaction, common omitted drivers, entry selection, or adaptive changes that jointly move both dimensions.

8.5 What empirical work would need

The formal structure points to the evidence required for an empirical test: independently defined measures of question significance and difficulty, observed entry into active inquiry, observed resolution and disengagement histories, and covariates that capture cross-effects such as resource mobilization or changing target conditions.

The model therefore distinguishes three empirical questions: whether difficult questions resolve more slowly, whether significant questions disengage more slowly, and whether cross-effects or prior dependence generate an observed relationship between significance and difficulty.

Potential empirical settings include biomedical research portfolios, mathematical open-problem databases, AI benchmark histories, and R&D project portfolios in which entry, continuation, abandonment, and completion can be observed.

9. Implications

The same observed state—unresolved—can be produced by different histories. A question may resist sustained inquiry, receive insufficient attention, face an identifiable informational or technical constraint, change as intervention proceeds, or generate a successor question. Management therefore requires distinguishing those histories rather than treating non-resolution as a single condition.

The distinction matters for research portfolios. A portfolio rich in difficult important questions can reflect successful removal of tractable questions, abandonment of low-value hard questions, persistent pursuit of consequential hard questions, or poor allocation that leaves important tractable questions underinvestigated.

It also matters for interpreting progress. A demanding unresolved population can coexist with successful inquiry if prior resolution removes easier questions or if progress opens new ones.

10. Limitations

Separate exit effects can enrich both significance and difficulty without inducing statistical dependence; stronger dependence requires cross-effects, prior association, or non-separable selection.

Question significance and difficulty are conceptually distinct but empirically difficult to measure. Applications will require domain-specific definitions rather than a universal scalar index.

Disengagement is rarely binary. Inquiry can move through low-intensity maintenance, dormancy, revival, and institutional abandonment. Entry into inquiry is also selected before the competing exit risks begin.

Empirically, disengagement is especially difficult to measure because low-intensity pursuit, dormancy, and abandonment can be hard to distinguish in observed records.

The selected dynamics are mechanism contrasts rather than causal tests. Empirical validation would require defined question populations, independent measures of significance and difficulty, and observed histories of entry, resolution, disengagement, and transformation.

11. Conclusion

The population of unresolved questions can become rich in both significance and difficulty even when the two properties are distinct.

Some questions leave because they are resolved. Others leave because meaningful pursuit stops. Difficulty can slow the first exit; significance can slow the second. These processes alter what remains visible. Statistical dependence between significance and difficulty requires additional coupling between the two processes or dependence already present at entry.

Capability growth adds further movement. Better information, methods, tools, and coordination can raise resolution rates, while progress can expose deeper questions and adaptive systems can change the target.

The visible set of unresolved questions is shaped by what we have answered, what we have stopped pursuing, and what still matters enough to keep trying.

Understanding why a question remains unresolved is therefore a prerequisite for deciding how to resolve it—or whether continued pursuit is warranted.

References

- [1] Carayol, N., & Dalle, J.-M. (2007). Sequential Problem Choice and the Reward System in Open Science. *Structural Change and Economic Dynamics*, 18(2), 167-191.
- [2] Frickel, S., Gibbon, S., Howard, J., Kempner, J., Ottinger, G., & Hess, D. J. (2010). Undone Science: Charting Social Movement and Civil Society Challenges to Research Agenda Setting. *Science, Technology, & Human Values*, 35(4), 444-473.

- [3] Posen, H. E., Keil, T., Kim, S., & Meissner, F. D. (2018). Renewing Research on Problemistic Search-A Review and Research Agenda. *Academy of Management Annals*, 12(1), 208-251.
- [4] Wrosch, C., Scheier, M. F., & Miller, G. E. (2023). Persistence and Disengagement in Personal Goal Pursuit. *Annual Review of Psychology*, 74.
- [5] Bloom, N., Jones, C. I., Van Reenen, J., & Webb, M. (2020). Are Ideas Getting Harder to Find? *American Economic Review*, 110(4), 1104-1144.
- [6] Elwert, F., & Winship, C. (2014). Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable. *Annual Review of Sociology*, 40, 31-53.
- [7] National Institute of Diabetes and Digestive and Kidney Diseases. Chronic Kidney Disease (CKD): causes, diagnosis, and management resources.
- [8] Clay Mathematics Institute. The Millennium Prize Problems; The Riemann Hypothesis.
- [9] Rosenbaum, D. I. (1993). Profit, Entry and Changes in Concentration. *International Journal of Industrial Organization*, 11(2), 185-203.
- [10] Stanford Institute for Human-Centered Artificial Intelligence. AI Index Report 2026. Technical Performance and Responsible AI sections.
- [11] Rzhetsky, A., Foster, J. G., Foster, I. T., & Evans, J. A. (2015). Choosing Experiments to Accelerate Collective Discovery. *Proceedings of the National Academy of Sciences*, 112(47), 14569-14574.
- [12] Alon, U. (2009). How to Choose a Good Scientific Problem. *Molecular Cell*, 35(6), 726-728.
- [13] McCardle, K. F., Tsetlin, I., & Winkler, R. L. (2018). When to Abandon a Research Project and Search for a New One. *Operations Research*, 66(3), 799-813.
- [14] Klingebiel, R., & Rammer, C. (2021). Optionality and Selectiveness in Innovation. *Academy of Management Discoveries*.
- [15] van Criekingen, K., Freel, M., & Czarnitzki, D. (2026). Open Innovation Deficiency: Evidence on Project Abandonment and Delay. *Journal of Technology Transfer*, 51, 1143-1180.
- [16] Linton, J. D., Walsh, S. T., & Morabito, J. (2002). Analysis, Ranking and Selection of R&D Projects in a Portfolio. *R&D Management*, 32(2), 139-148.